

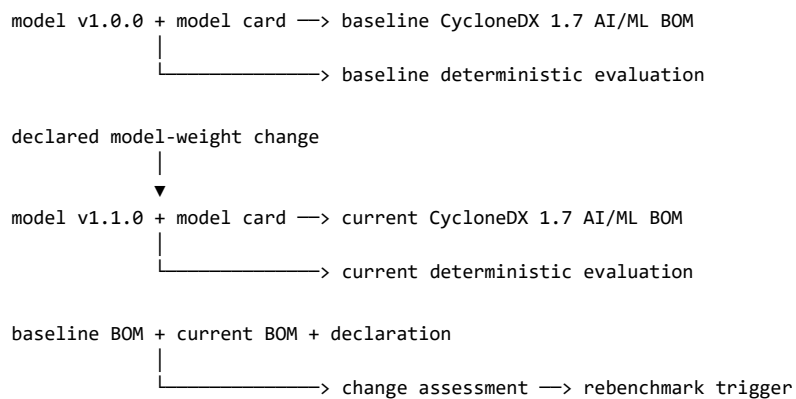
AI/ML BOM Model-Change Demonstration

Purpose and scope

This package demonstrates a machine-checkable model-change workflow for a synthetic ONNX model. It is evidence that the proposed mechanism can be built with current tools; it is not clinical evidence and is not an FDA submission for a medical device.

The fixture contains no patient data. The model is handcrafted, is not clinically validated, and must not be used for diagnosis, treatment, patient care, or clinical decision support.

Evidence flow



Demonstrated change

Field	Baseline	Current
Model version	1.0.0	1.1.0
ONNX SHA-256	17e341bde16199eea432a840f27f071cb30baa7a06314e1eea4427d2b52ea59c	fb30420e66ab9850a7d740b29bdc04f0c1bc92a325d71156b4f230aadb99e248
Declared behavior class	—	model_weights
Evaluation result	3/3 exact sequences	3/3 exact sequences

The content hash and semantic version changed while the documented output

contract remained stable on the fixed synthetic evaluation set. The generated assessment confirms that the declaration matches the observed BOM difference.

The deterministic trigger requires baseline-performance, model-integrity, and safety-regression suites.

Review the artifacts

1. Compare the machine-learning-model component in

artifacts/baseline/device-sbom.cdx.json and

artifacts/current/device-sbom.cdx.json.

2. Review the supplier declaration in

changes/model-change-declaration.json.

3. Review the observed change and declaration consistency result in

artifacts/model-change-assessment.json.

4. Review the deterministic decision and required suites in

artifacts/rebenchmark-trigger-report.json.

5. Review the complete generation provenance in

artifacts/generation-manifest.json.

6. Run python validators/validate_artifacts.py to validate schemas, hashes,

evaluation equivalence, evidence cross-references, and SHA256SUMS.

Reproduce and verify

```
python -m venv .venv
.venv\Scripts\python -m pip install -r requirements.txt
.venv\Scripts\python model\generate_model.py
.venv\Scripts\python scripts\evaluate.py --check
.venv\Scripts\python -m pytest -q
.venv\Scripts\python validators\validate_artifacts.py
```

The exact SBOMator generation command and tool revisions are recorded in

artifacts/generation-manifest.json. CycloneDX serial numbers and metadata

timestamps are unique per run; remove those fields before byte-level comparison.

Format boundaries and limitations

- The AI/ML BOMs conform to CycloneDX 1.7 and include the model as a machine-learning-model component with a model card, version, supplier, license, and content hash.
- The conventional vulnerability VEX documents conform to CycloneDX 1.6 and contain no vulnerability statements because online vulnerability lookup was disabled for this deterministic fixture.
- The model-change declaration, assessment, and rebenchmark trigger use open project JSON Schemas. They are inspired by VEX's machine-readable component/status/analysis pattern but are not represented as CycloneDX VEX vulnerability statements.
- Stable output on three synthetic examples does not establish model safety, efficacy, generalization, or clinical performance.